Student Name: _____

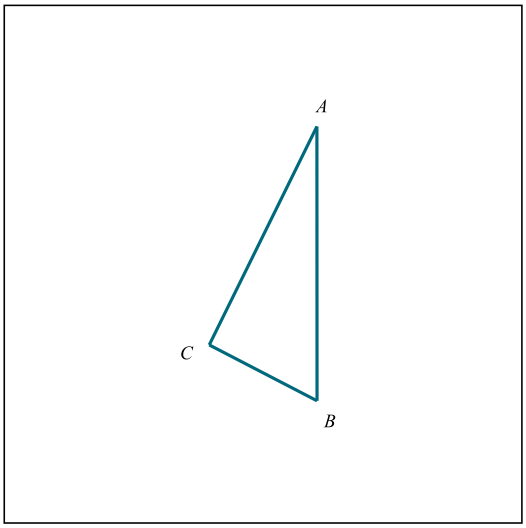
Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **48**

Instructor Name: **Bodanske, Daniel**

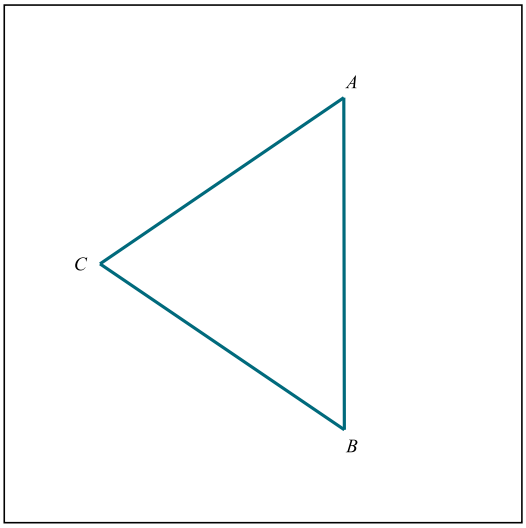
Question 1 of 48

Use a compass and ruler to circumscribe a circle about the triangle.



Question 2 of 48

Use a compass and ruler to inscribe a circle in the triangle.



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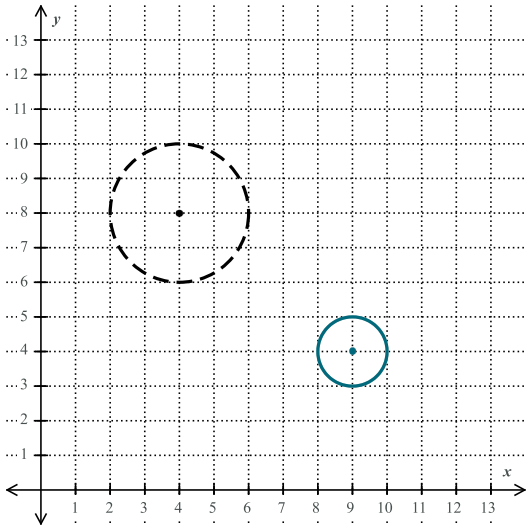
Question 3 of 48

In the figure below, the solid circle has center $(9,\ 4)$ and radius 1.
The dashed circle has center $(4,\ 8)$ and radius 2.

Use the transformation tools given to move the solid circle exactly onto the dashed circle.
Then answer the parts below.



(a) Fill in the blanks to describe the transformations necessary to move the solid circle exactly onto the dashed circle.

Translate the solid circle {to the left, to the right} by _____ unit(s) and {up, down} by _____ unit(s).

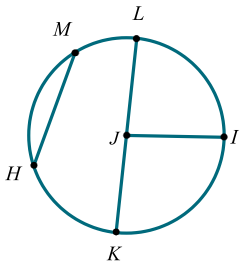
Dilate the solid circle by a scale factor of _____.

(b) Are the original solid circle and the dashed circle similar?

- ○ Yes
- ○ No

Question 4 of 48

A circle with center J is shown in the figure below.



(a) Name a radius

(b) Name a diameter

(c) Name a chord

(d) If the length of $\overline{JI}$ is 3 units,
what is the length of $\overline{KL}$?

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Question 5 of 48

The circumference of a circular garden is 97.34 feet. What is the radius of the garden? Use 3.14 for π and do not round your answer.

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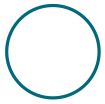
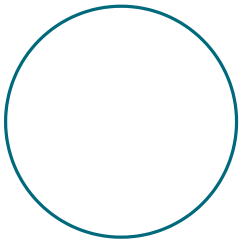
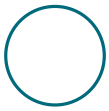
Question 6 of 48

Below are three circles.
The circumference and diameter of each circle have been approximately measured.

Circle 1:	**Circle 2:**	**Circle 3:**
Circumference: 23.565 units	Circumference: 58.92 units	Circumference: 25.134 units
Diameter: 7.5 units	Diameter: 18.75 units	Diameter: 8 units
		

(a) Use the measurements above and a calculator to fill in the blanks.
Write each answer as a single decimal. Do not round.

Circle 1: $\frac{\text{Circumference}}{\text{Diameter}} =$

Circle 2: $\frac{\text{Circumference}}{\text{Diameter}} =$

Circle 3: $\frac{\text{Circumference}}{\text{Diameter}} =$

(b) Let C represent the circumference of any circle and d represent the diameter.

○ $\frac{d}{C} = \pi$ ○ $\frac{d}{\pi} = C$ ○ $\frac{\pi}{C} = d$ ○ $\frac{C}{d} = \pi$

(c) Write a formula that relates the circumference, C, of a circle to its diameter, d.

$C =$

(a) Use the measurements above and a calculator to fill in the blanks.
Write each answer as a single decimal. Do not round.

Circle 1: $\frac{\text{Circumference}}{\text{Diameter}} =$ _____

Circle 2: $\frac{\text{Circumference}}{\text{Diameter}} =$ _____

Circle 3: $\frac{\text{Circumference}}{\text{Diameter}} =$ _____

(b) Let C represent the circumference of any circle and d represent the diameter.
Using the fact that π is approximately 3.14, which formula seems reasonable?

○ $\frac{d}{C} = \pi$

○ $\frac{d}{\pi} = C$

○ $\frac{\pi}{C} = d$

○ $\frac{C}{d} = \pi$

(c) Write a formula that relates the circumference, C, of a circle to its diameter, d.

$C =$ _____

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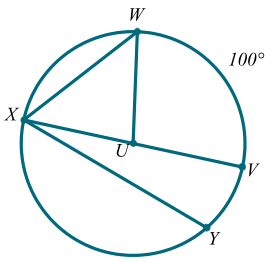
Question 7 of 48

In the circle below, U is the center, $\overline{VX}$ is a diameter, and $m\overset{\frown}{VW} = 100°$. Use this information to find the following.

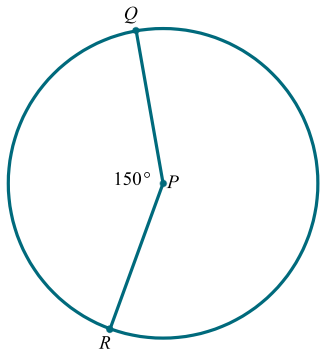


(a) Give a central angle
(b) Give a major arc
(c) Give a minor arc
(d) Find $m\angle VUW$
(e) Find $m\overset{\frown}{VYW}$

Question 8 of 48

The circle below has center P, and its radius is 5 yd. Given that $m\angle QPR = 150°$, find the length of the minor arc $\overset{\frown}{QR}$.

Give an exact answer in terms of π, and be sure to include the correct unit in your answer.



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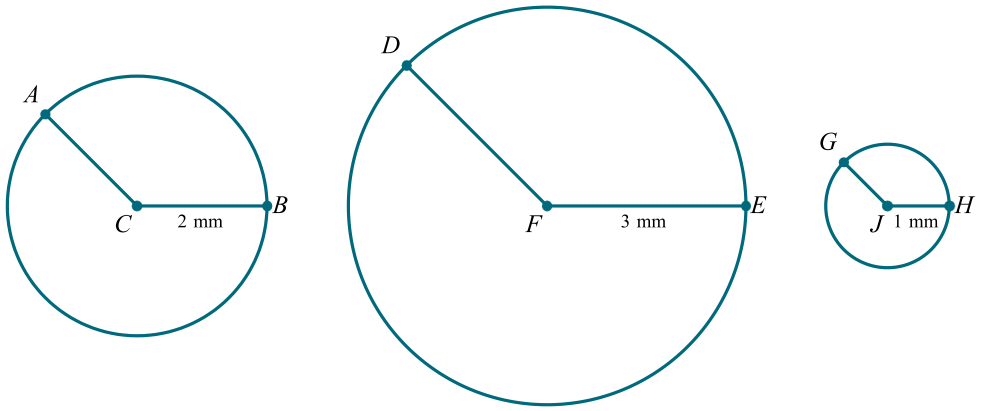
Question 9 of 48

Below are circles C, F, and J.
A central angle of $135°$ is shown in each.



Answer the following.
When applicable, give the exact answer in terms of π in simplest form.

(a) For each circle, find the length of the arc that is intercepted by the central angle.

Length of $\overset{\frown}{AB}$: _____ mm

Length of $\overset{\frown}{DE}$: _____ mm

Length of $\overset{\frown}{GH}$: _____ mm

(b) For each circle, find the ratio of the arc length to the radius.

$$\frac{\text{Length of } \overset{\frown}{AB}}{\text{Radius of circle } C} =$$

$$\frac{\text{Length of } \overset{\frown}{DE}}{\text{Radius of circle } F} =$$

$$\frac{\text{Length of } \overset{\frown}{GH}}{\text{Radius of circle } J} =$$

(c) The ratios suggest which of the following?

- ○ The length of the arc intercepted by a central angle is $\frac{3\pi}{4}$ units more than the radius.
- ○ The ratios increase by one unit as the lengths of the intercepted arcs increase by $\frac{3\pi}{4}$ units.
- ○ The length of the arc intercepted by a central angle is proportional to the radius.
- ○ The ratios increase by $\frac{3\pi}{4}$ units as the radii increase by one unit.

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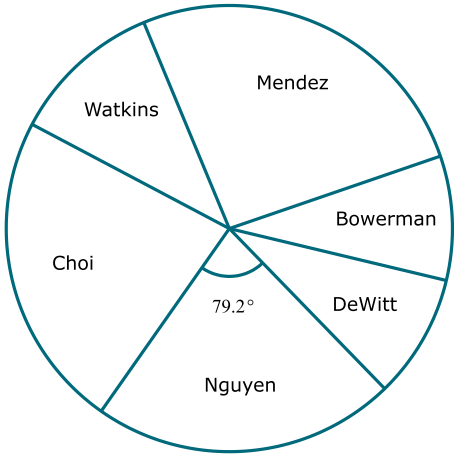
Question 10 of 48

In a recent election, there were six candidates and 5100 total votes. The circle graph below summarizes the votes by candidate.

The central angle for the Nguyen slice is $79.2°$. How many of the votes did Nguyen get?



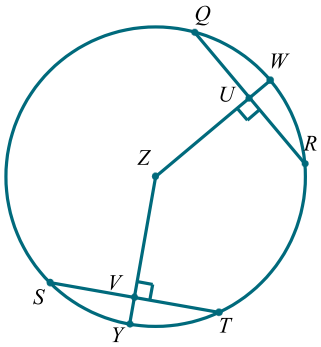
Question 11 of 48

Convert $600°$ to radian measure in terms of π.

[ ] radians

Question 12 of 48

In the figure below, Z is the center of the circle. Suppose that $ST = 18$, $RU = 9$, $QU = 4x - 3$, and $UZ = 5$. Find the following.



$x =$ [ ]

$VZ =$ [ ]

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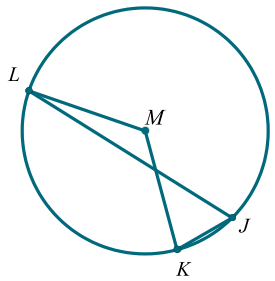
Question 13 of 48

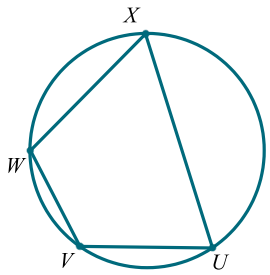
The circle below has center M. Suppose that $m\angle KJL = 62°$. Find the following.

(a) $m\overset{\frown}{KL}$
(b) $m\angle KML$



Question 14 of 48

In the circle below, suppose $m\overset{\frown}{VUX} = 216°$ and $m\angle UVW = 118°$. Find the following.



(a) $m\angle VUX =$ _____°

(b) $m\angle UXW =$ _____°

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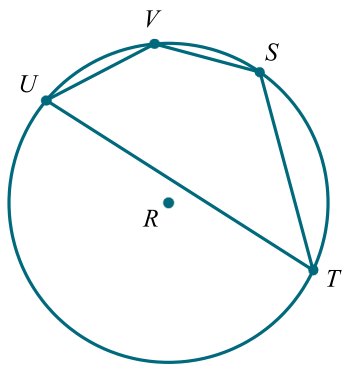
Question 15 of 48

Quadrilateral $STUV$ is inscribed in circle R. Complete the following.



(a) Choose the correct expression to finish each statement.

$$m\angle T = \{2m\overset{\frown}{STU},\ \frac{1}{2}m\overset{\frown}{STU},\ \frac{1}{2}m\overset{\frown}{TUV},\ m\overset{\frown}{TUV},\ 2m\overset{\frown}{UVS},\ \frac{1}{2}m\overset{\frown}{UVS},\ m\overset{\frown}{VST},\ \frac{1}{2}m\overset{\frown}{VST}\}$$

$$m\angle V = \{2m\overset{\frown}{STU},\ \frac{1}{2}m\overset{\frown}{STU},\ \frac{1}{2}m\overset{\frown}{TUV},\ m\overset{\frown}{TUV},\ 2m\overset{\frown}{UVS},\ \frac{1}{2}m\overset{\frown}{UVS},\ m\overset{\frown}{VST},\ \frac{1}{2}m\overset{\frown}{VST}\}$$

(b) Choose the correct expressions for the sum of $m\angle T$ and $m\angle V$.

$$\begin{aligned} m\angle T + m\angle V = \{&\frac{1}{2}m\overset{\frown}{TUV} + \frac{1}{2}m\overset{\frown}{VST},\ m\overset{\frown}{TUV} + m\overset{\frown}{VST},\\ &\frac{1}{2}m\overset{\frown}{UVS} + \frac{1}{2}m\overset{\frown}{STU},\ 2m\overset{\frown}{UVS} + 2m\overset{\frown}{STU},\\ &\frac{1}{2}m\overset{\frown}{VST} + \frac{1}{2}m\overset{\frown}{STU},\ m\overset{\frown}{VST} + m\overset{\frown}{STU},\\ &m\overset{\frown}{TUV} + m\overset{\frown}{UVS},\ \frac{1}{2}m\overset{\frown}{TUV} + \frac{1}{2}m\overset{\frown}{UVS}\}\\ = \{&\frac{1}{2}\left(m\overset{\frown}{TUV} + m\overset{\frown}{VST}\right),\ 2m\overset{\frown}{TUV} + 2m\overset{\frown}{VST},\\ &\frac{1}{2}\left(m\overset{\frown}{UVS} + m\overset{\frown}{STU}\right),\ 2\left(m\overset{\frown}{UVS} + m\overset{\frown}{STU}\right),\\ &\frac{1}{2}\left(m\overset{\frown}{VST} + m\overset{\frown}{STU}\right),\ 2\left(m\overset{\frown}{VST} + m\overset{\frown}{STU}\right),\\ &2m\overset{\frown}{TUV} + 2m\overset{\frown}{UVS},\ \frac{1}{2}\left(m\overset{\frown}{TUV} + m\overset{\frown}{UVS}\right)\} \end{aligned}$$

(c) Fill in the blank with the correct number.

$m\angle T + m\angle V =$ _____ °

(d) From the steps above, classify $\angle T$ and $\angle V$.

- ○ $\angle T$ and $\angle V$ are supplementary.
- ○ $\angle T$ and $\angle V$ are complementary.
- ○ $\angle T$ and $\angle V$ are neither supplementary nor complementary.

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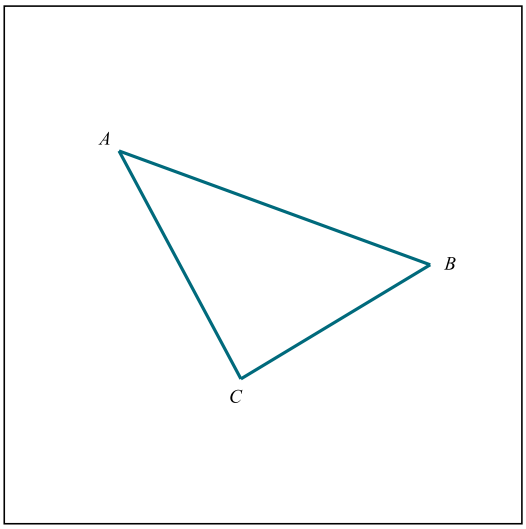
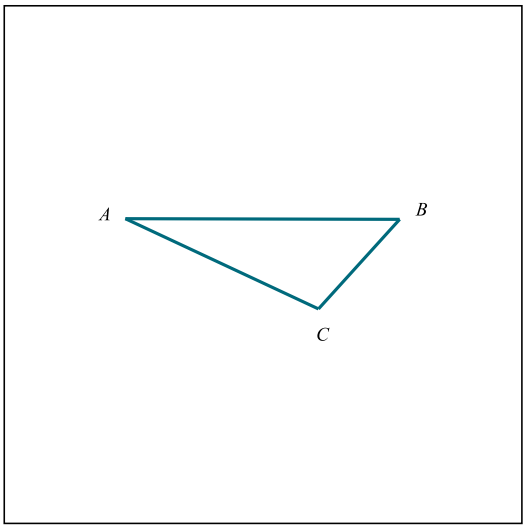
Question 16 of 48

Use a compass and ruler to inscribe a circle in the triangle.



Question 17 of 48

Use a compass and ruler to circumscribe a circle about the triangle.



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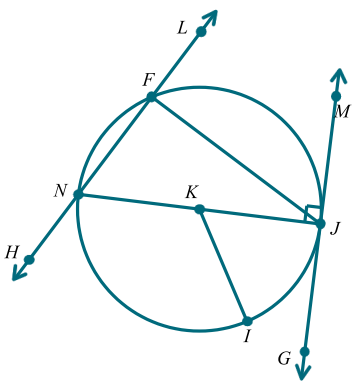
Question 18 of 48

The circle below has center K and diameter $\overline{NJ}$. $\overleftrightarrow{MG}$ intersects the circle at J, and $\overleftrightarrow{LH}$ intersects the circle at F and N. Use this information to find the following.



(a) Give a secant line.

$\overleftrightarrow{}$

$\overline{}$

(b) Give a chord.

$\overline{}$

$\overline{}$

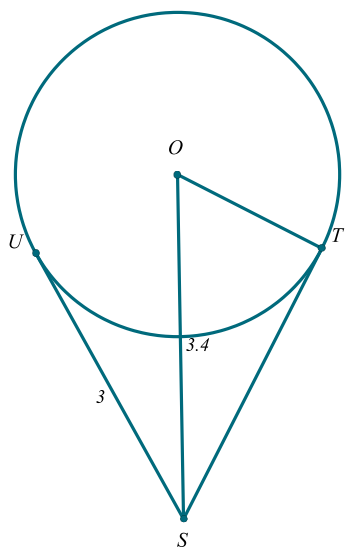
(c) Give a tangent line.

$\overleftrightarrow{}$

$\overline{}$

Question 19 of 48

In the figure below, the segments $\overline{ST}$ and $\overline{SU}$ are tangent to the circle centered at O. Given that $SU = 3$ and $OS = 3.4$, find OT.



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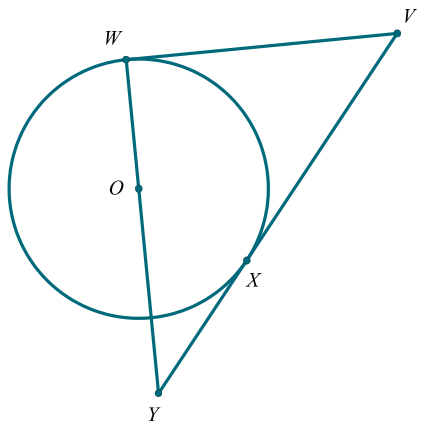
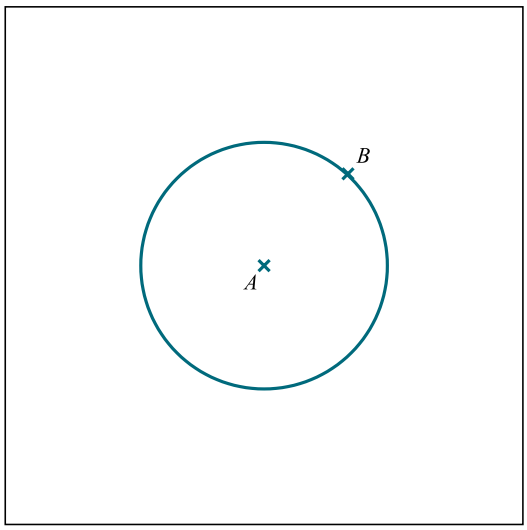
Question 20 of 48

In the figure below, $\overline{VY}$ and $\overline{VW}$ are tangent to the circle centered at O. The segment $\overline{YW}$ passes through the center of the circle. Given that $VY = 14.5$ and $YW = 10$, find YX.



Question 21 of 48

In the given figure, A is the center of the circle. Use a compass and ruler to construct a tangent line for the given circle that passes through point B.



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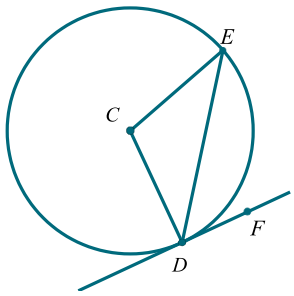
Question 22 of 48

The circle below has center C. Suppose that $m\overset{\frown}{DE} = 106°$ and that $\overline{DF}$ is tangent to the circle at D. Find the following.

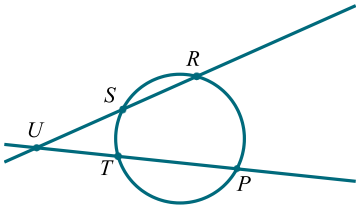
(a) $m\angle EDF$
(b) $m\angle DCE$



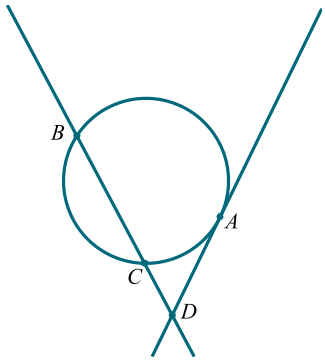
Question 23 of 48

Find the missing information for both parts below.

(a) In the figure below, $m\angle PUR = 30°$ and $m\overset{\frown}{PR} = 103°$. Find $m\overset{\frown}{ST}$.



(b) In the figure below, $m\angle ADC = 54°$ and $m\overset{\frown}{AC} = 65°$. Find $m\overset{\frown}{AB}$.



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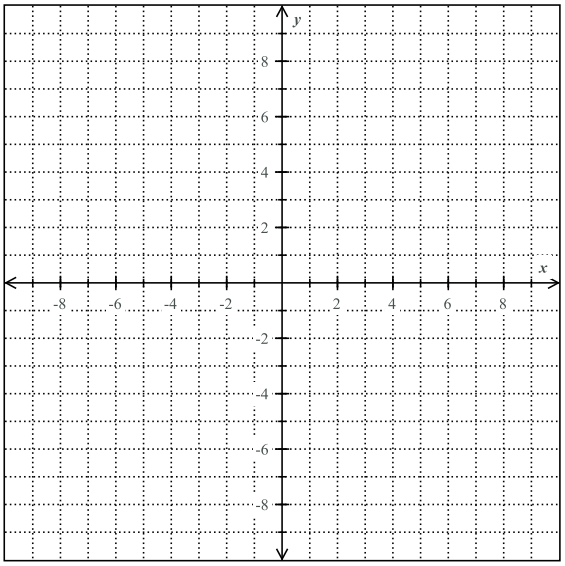
Question 24 of 48

The equation of a circle is given below. Identify the radius and center. Then graph the circle.

$$x^2 + (y-1)^2 = 4$$

Radius: _____

Center: (_____, _____)



Question 25 of 48

Give the equation of the circle centered at the origin and passing through the point $(-4,\ 0)$.

Question 26 of 48

Consider the circle centered at the origin and passing through the point $(4,\ 0)$.

(a) Give the equation of the circle.

(b) For each point, decide whether or not it is on the circle.

$(x,\ y)$	**Is the point on the circle?**	
	Yes	**No**
$(-1,\ \sqrt{7})$	○	○
$(0,\ 5)$	○	○
$(-4,\ 0)$	○	○
$(\sqrt{5},\ 2)$	○	○

Question 27 of 48

Write an equation of the circle with center $(-2,\ -3)$ and radius 11.

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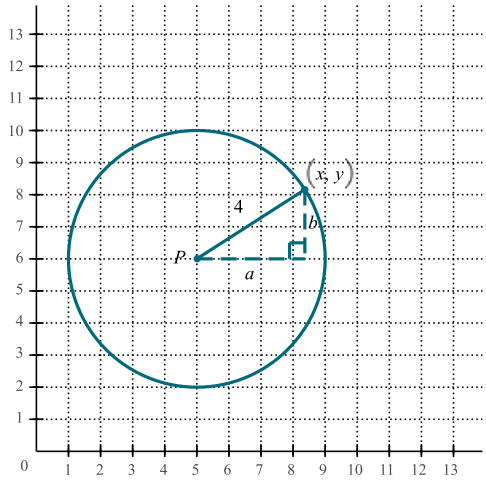
Question 28 of 48

The circle below has center P.
The point (x, y) is on the circle as shown.



(a) Find the following.

Center: (_____, _____)

Radius: _____ units

Value of a: $\{y-6,\ 6-y,\ x-5,\ 5-x,\ 5-y,\ 6-x,\ y-5,\ x-6\}$

Value of b: $\{y-6,\ 6-y,\ x-5,\ 5-x,\ 5-y,\ 6-x,\ y-5,\ x-6\}$

(b) Use the Pythagorean Theorem to write an equation relating the side lengths of the right triangle. Leave your answer in terms of x and y.

$(_____)^2 + (_____)^2 = (_____)^2$

Question 29 of 48

Find an equation of the circle that has center $(0, 2)$ and passes through $(-3, -3)$.

Question 30 of 48

Find an equation of the circle whose diameter has endpoints $(-5, 1)$ and $(-1, -2)$.

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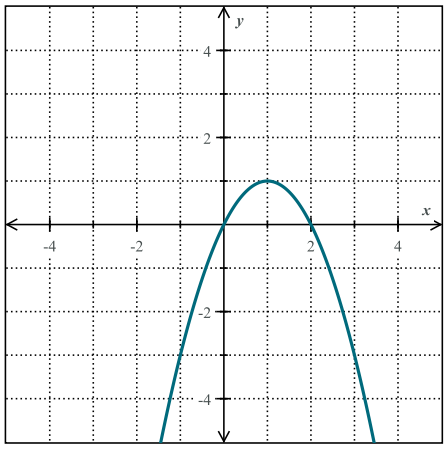
Question 31 of 48

The graph of a function is given below.
Give all x-intercepts and y-intercepts shown.



If there is more than one answer, separate them with commas.

Click on "None" if applicable.

x-intercept(s): _____

y-intercept(s): _____

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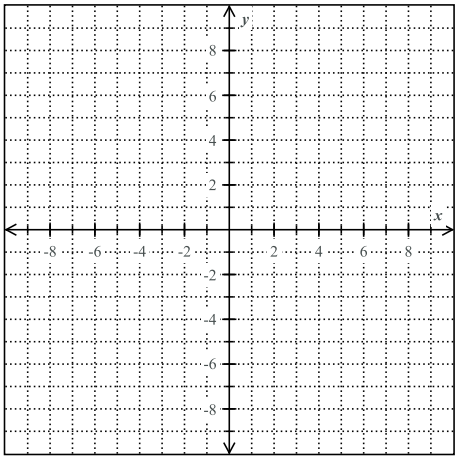
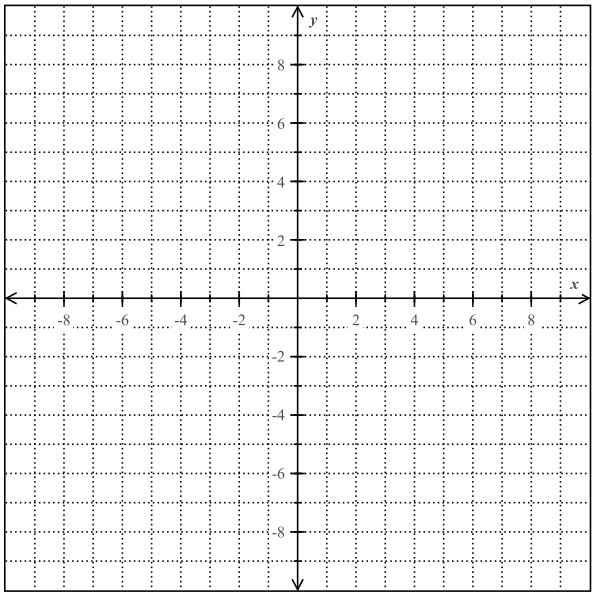
Question 32 of 48

Graph the parabola.

$$y = (x+4)^2 - 3$$





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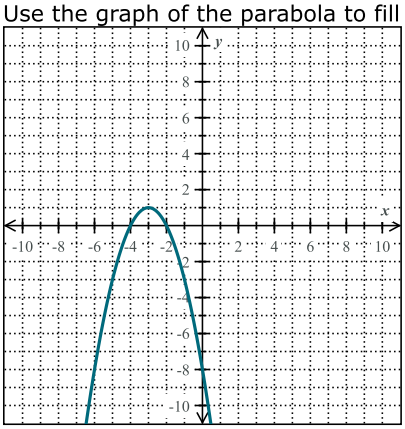
Question 33 of 48

Use the graph of the parabola to fill in the table.



(a) Does the parabola open upward or downward?

upward downward

(b) Find the equation of the axis of symmetry.

equation of axis of symmetry: $\square$

(c) Find the intercept(s).

x-intercept(s): $\square$

y-intercept(s): $\square$

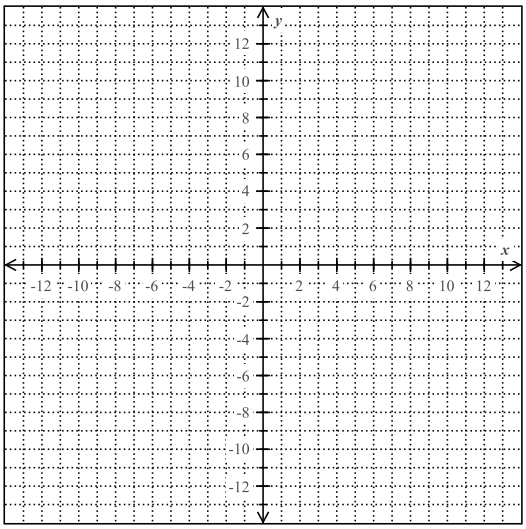
(d) Find the coordinates of the vertex.

vertex: $(\square, \square)$

Question 34 of 48

Graph the parabola.

$$y = -3(x+4)^2 - 1$$



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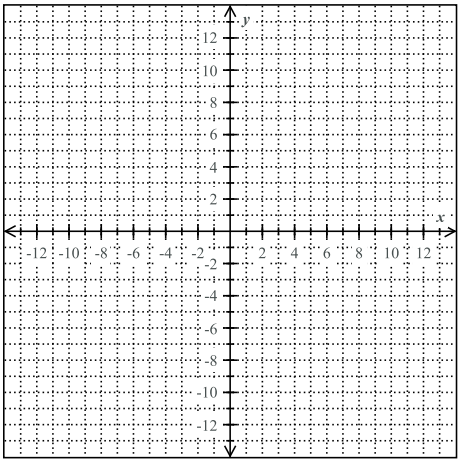
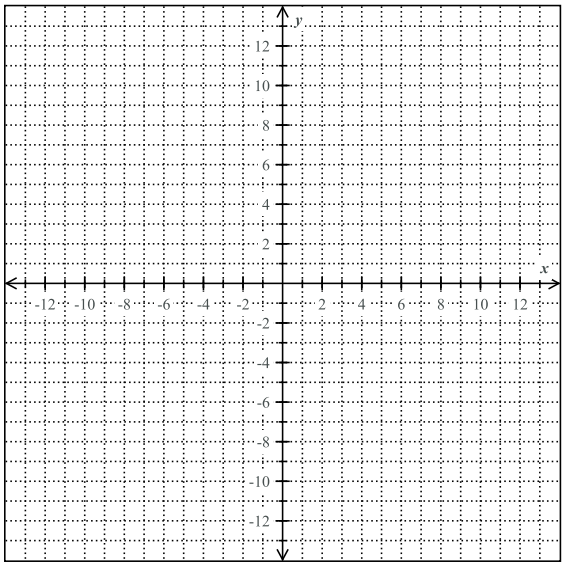
Question 35 of 48

Graph the parabola.

$$y = x^2 + 10x + 26$$





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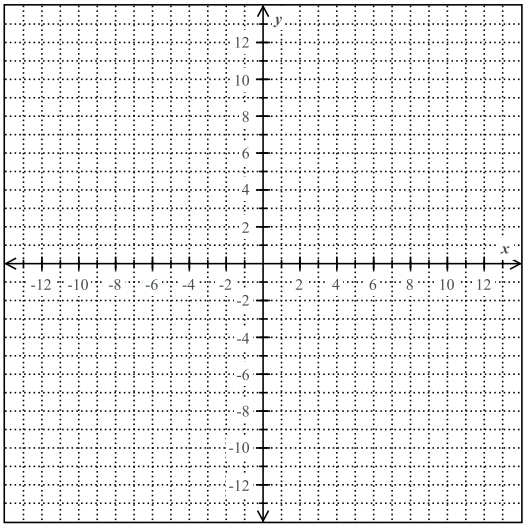
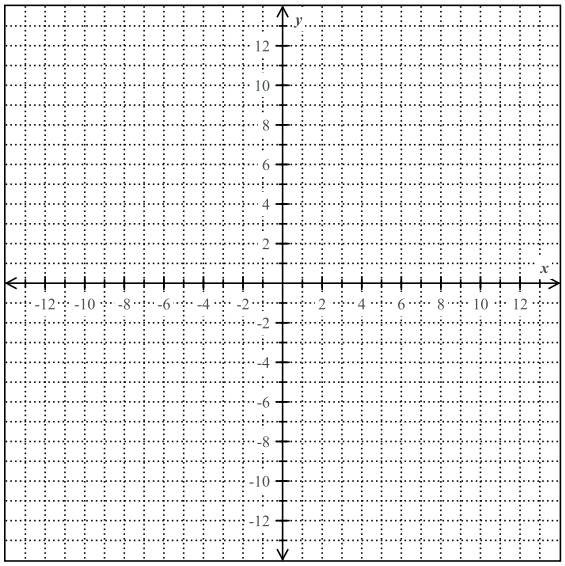
Question 36 of 48

Graph the parabola.

$$y^2 = \frac{1}{4}x$$





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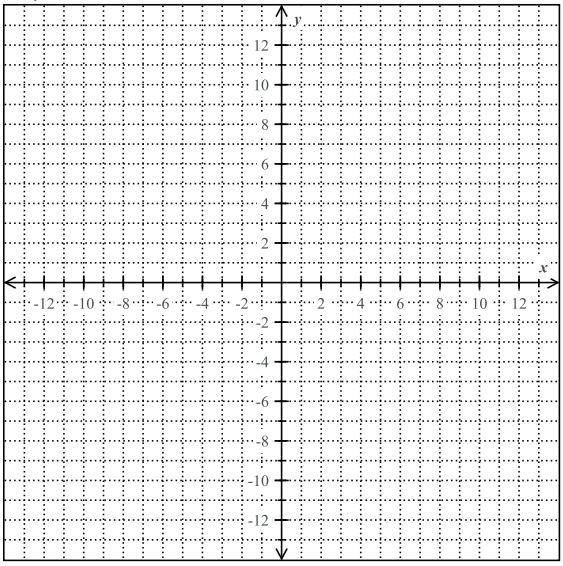
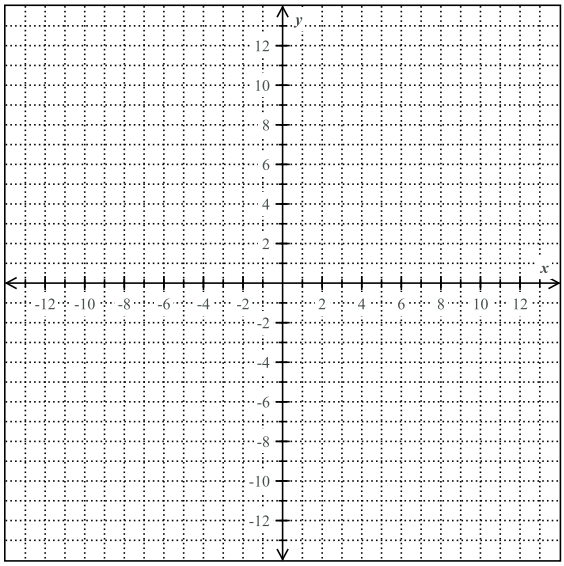
Question 37 of 48

Graph the parabola.

$$y = -3(x+5)^2 + 1$$





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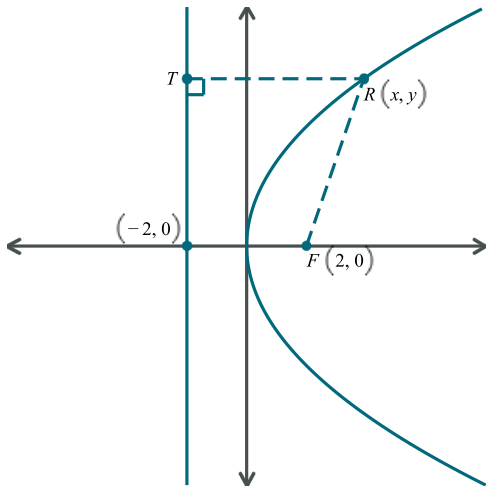
Question 38 of 48

The parabola in the figure below has vertex $(0, 0)$ and focus $F(2, 0)$.
Its directrix is the vertical line that passes through $(-2, 0)$.
The point $R(x, y)$ is a point on the parabola that is above and to the right of the focus. The point T lies on the directrix, and $\overline{RT}$ is perpendicular to the directrix.



(a) What are the coordinates for the point T ? $\{(-2, 0), (2, y), (2, 0), (x, y), (-2, y), (x, 0)\}$
(b) Write the expressions for the distances FR and TR. FR = _____ TR = _____
(c) From the definition of a parabola, which of the following *must* be true? $\{FR = TR, FR + TR = 0, FR < TR, FR > TR\}$
(d) Use the above to write an equation for the given parabola solved for y^2. y^2 = _____

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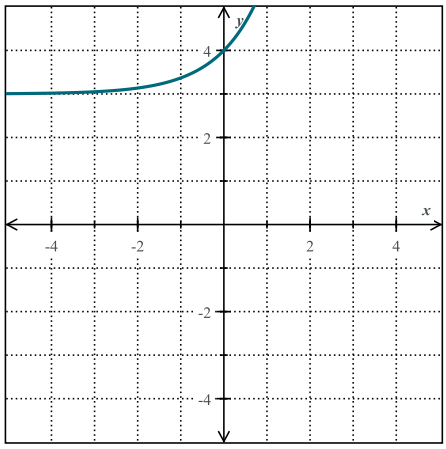
Question 39 of 48

The graph of a function is given below.
Give all y-intercepts and x-intercepts shown.



If there is more than one answer, separate them with commas.

Click on "None" if applicable.

y-intercept(s): _____

x-intercept(s): _____

Question 40 of 48

Simplify.

$$(4v)^3$$

Write your answer without parentheses.

Question 41 of 48

Simplify.

$$\left(-7u^2+2u-5\right)+\left(7u^2+6u\right)$$

Question 42 of 48

Multiply.

$$(w-8)(w+1)$$

Simplify your answer.

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Question 43 of 48

Multiply.

$$(3a-1)(a-6)$$

Simplify your answer.

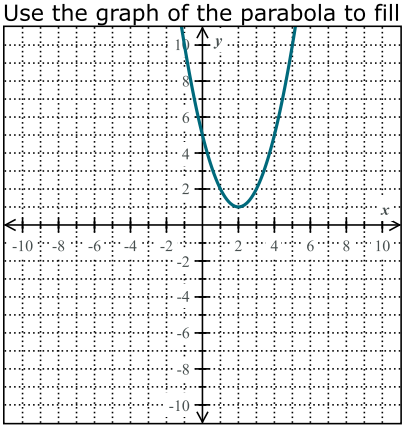
Question 44 of 48

Rewrite without parentheses and simplify.

$$(2-3u)^2$$

Question 45 of 48

Use the graph of the parabola to fill in the table.



(a) Does the parabola open upward or downward?

upward downward

(b) Find the equation of the axis of symmetry.

equation of axis of symmetry: ☐

(c) Find the coordinates of the vertex.

vertex: (☐, ☐)

(d) Find the intercept(s).

x-intercept(s): ☐

y-intercept(s): ☐

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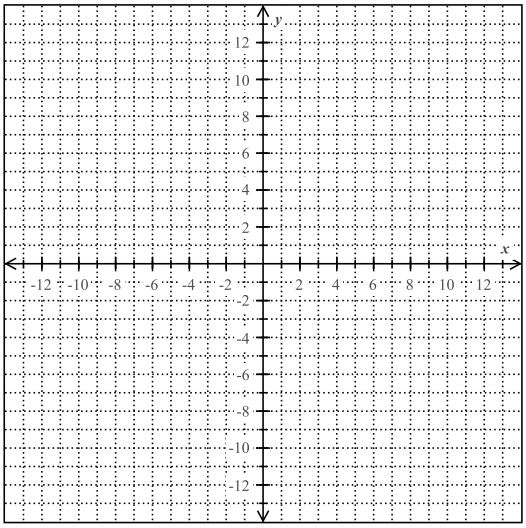
Question 46 of 48

Graph the parabola.

$$y = 2(x+6)^2 + 4$$



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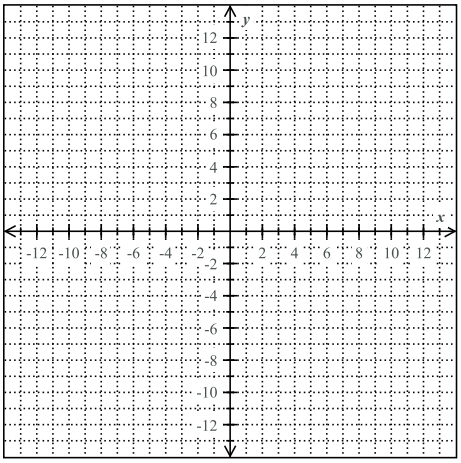
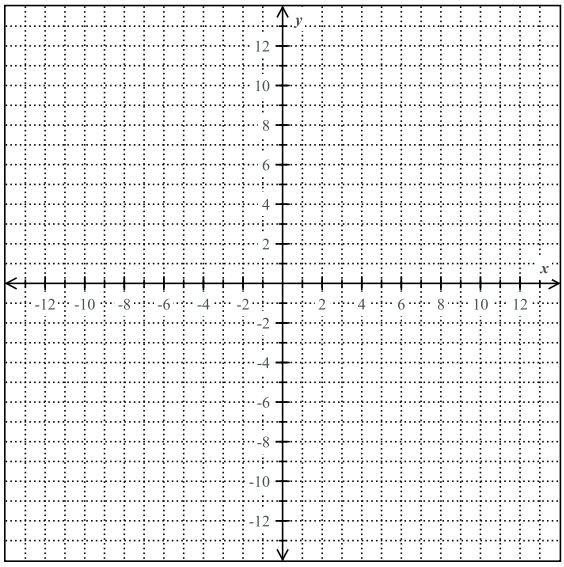
Question 47 of 48

Graph the parabola.

$$y = x^2 - 6x + 14$$





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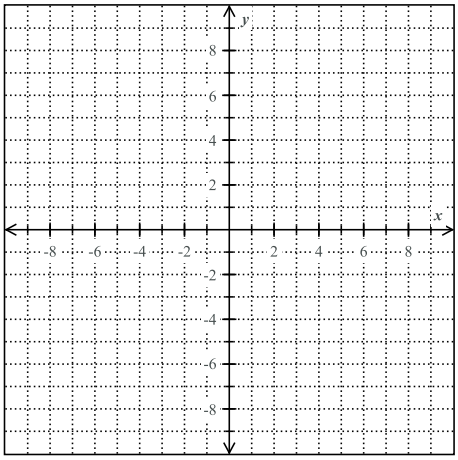
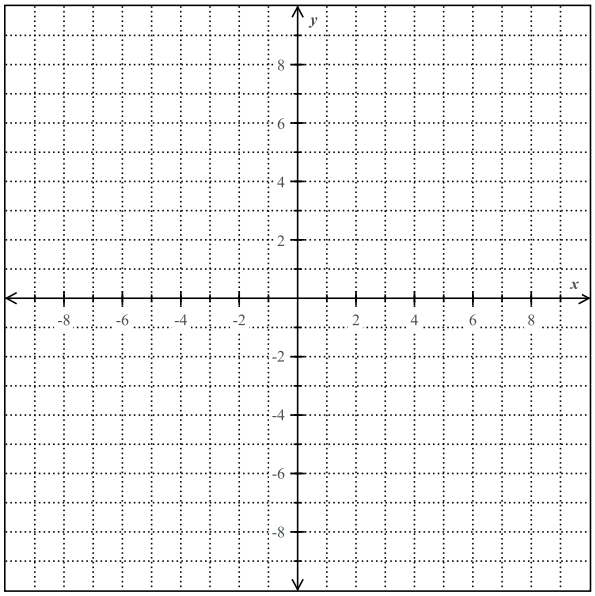
Question 48 of 48

Graph the parabola.

$$y = (x-1)^2 + 4$$





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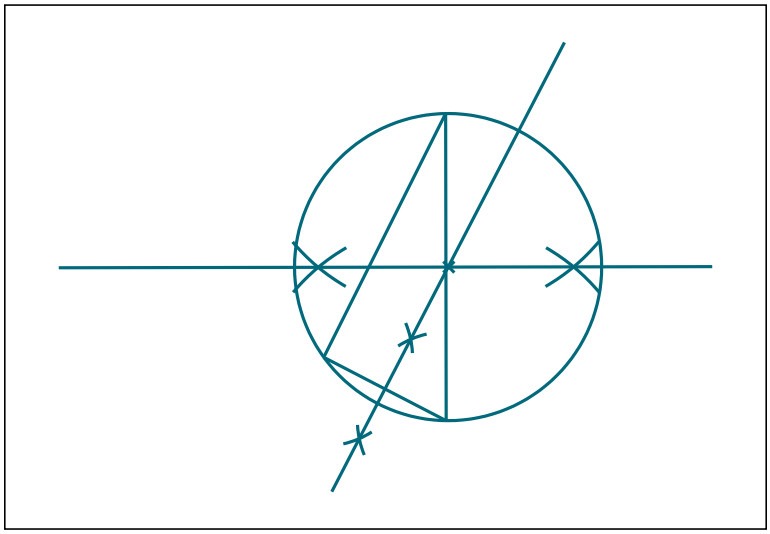
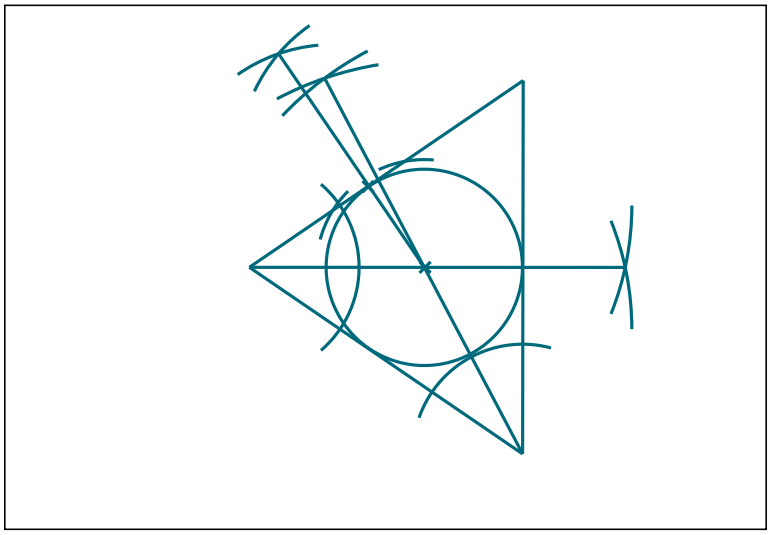
Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **48**

Question 1 of 48



Question 2 of 48



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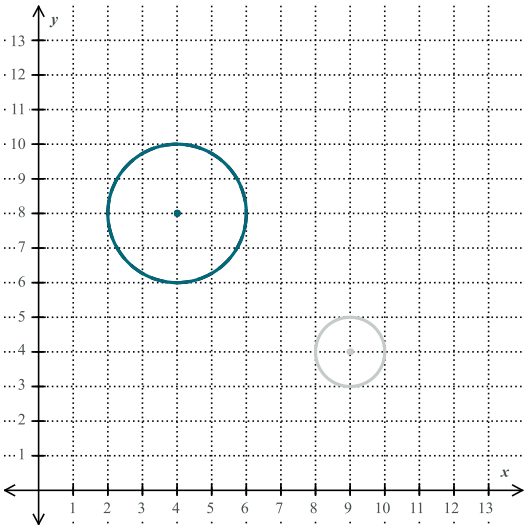
Question 3 of 48



(a) Fill in the blanks to describe the transformations necessary to move the solid circle exactly onto the dashed circle.

Translate the solid circle [to the left] by 5 unit(s) and [up] by 4 unit(s).

Dilate the solid circle by a scale factor of 2.

(b) Are the original solid circle and the dashed circle similar?

- (●) Yes
- ( ) No

Question 4 of 48

(a) Name a radius: $\overline{JK}$

(b) Name a diameter: $\overline{KL}$

(c) Name a chord: $\overline{HM}$

(d) If the length of $\overline{JI}$ is 3 units,

what is the length of $\overline{KL}$? 6 units

Question 5 of 48

15.5 feet

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Question 6 of 48

(a) **Circle 1:**

$$\frac{\text{Circumference}}{\text{Diameter}} = 3.142$$

Circle 2:

$$\frac{\text{Circumference}}{\text{Diameter}} = 3.1424$$

Circle 3:

$$\frac{\text{Circumference}}{\text{Diameter}} = 3.14175$$

(b)

- ○ $\frac{d}{C} = \pi$
- ○ $\frac{d}{\pi} = C$
- ○ $\frac{\pi}{C} = d$
- ◉ $\frac{C}{d} = \pi$

(c) $C = \pi d$

Question 7 of 48

(a) Give a central angle: $\angle WUV$

(b) Give a major arc: $\overset{\frown}{VXW}$

(c) Give a minor arc: $\overset{\frown}{VW}$

(d) Find $m\angle VUW$: $100°$

(e) Find $m\overset{\frown}{VYW}$: $260°$

Question 8 of 48

Length of minor arc $\overset{\frown}{QR}$: $\frac{25}{6}\pi$ yd

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Question 9 of 48

(a) For each circle, find the length of the arc that is intercepted by the central angle.

Length of $\overset{\frown}{AB}$: $\frac{3\pi}{2}$ mm

Length of $\overset{\frown}{DE}$: $\frac{9\pi}{4}$ mm

Length of $\overset{\frown}{GH}$: $\frac{3\pi}{4}$ mm

(b) For each circle, find the ratio of the arc length to the radius.

$\frac{\text{Length of } \overset{\frown}{AB}}{\text{Radius of circle } C} = \frac{3\pi}{4}$

$\frac{\text{Length of } \overset{\frown}{DE}}{\text{Radius of circle } F} = \frac{3\pi}{4}$

$\frac{\text{Length of } \overset{\frown}{GH}}{\text{Radius of circle } J} = \frac{3\pi}{4}$

(c) The ratios suggest which of the following?

- ○ The length of the arc intercepted by a central angle is $\frac{3\pi}{4}$ units more than the radius.
- ○ The ratios increase by one unit as the lengths of the intercepted arcs increase by $\frac{3\pi}{4}$ units.
- ◉ The length of the arc intercepted by a central angle is proportional to the radius.
- ○ The ratios increase by $\frac{3\pi}{4}$ units as the radii increase by one unit.

Question 10 of 48

1122

Question 11 of 48

$\frac{10\pi}{3}$ radians

Question 12 of 48

$x = 3$

$VZ = 5$

Question 13 of 48

(a) $m\overset{\frown}{KL} = 124°$

(b) $m\angle KML = 124°$

Question 14 of 48

(a) $m\angle VUX = 72°$
(b) $m\angle UXW = 62°$

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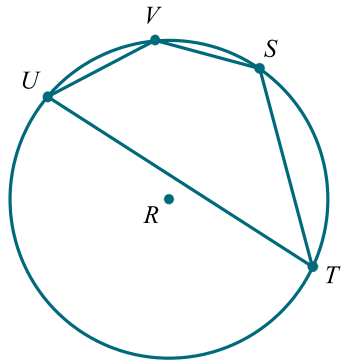
Question 15 of 48



(a) Choose the correct expression to finish each statement.

$m\angle T = \frac{1}{2} m\widehat{UVS}$

$m\angle V = \frac{1}{2} m\widehat{STU}$

(b) Choose the correct expressions for the sum of $m\angle T$ and $m\angle V$.

$m\angle T + m\angle V = \frac{1}{2} m\widehat{UVS} + \frac{1}{2} m\widehat{STU}$

$= \frac{1}{2}\left(m\widehat{UVS} + m\widehat{STU}\right)$

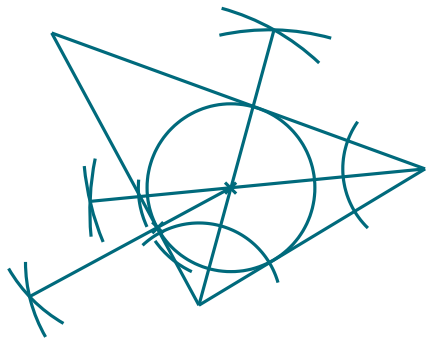
(c) Fill in the blank with the correct number.

$m\angle T + m\angle V = 180°$

(d) From the steps above, classify $\angle T$ and $\angle V$.

- (●) $\angle T$ and $\angle V$ are supplementary.
- ( ) $\angle T$ and $\angle V$ are complementary.
- ( ) $\angle T$ and $\angle V$ are neither supplementary nor complementary.

Question 16 of 48



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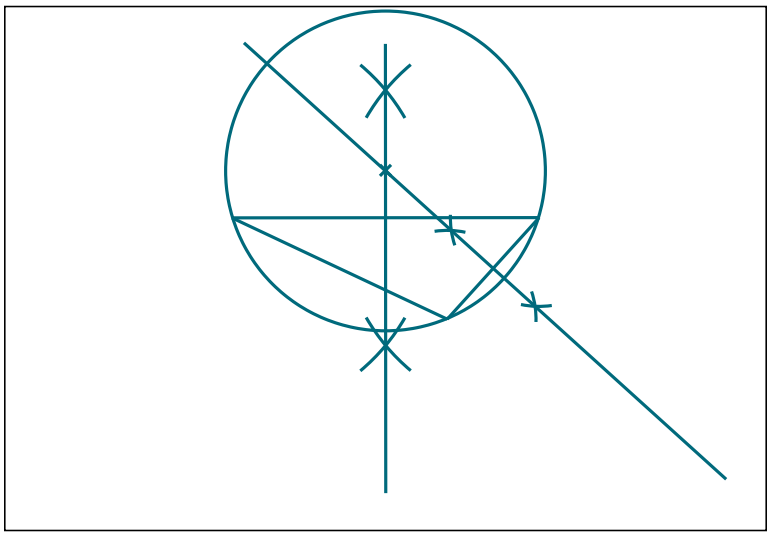
Question 17 of 48



Question 18 of 48

(a) Give a secant line.

$\overleftrightarrow{LH}$

(b) Give a chord.

$\overline{FJ}$

(c) Give a tangent line.

$\overleftrightarrow{MG}$

Question 19 of 48

$OT = 1.6$

Question 20 of 48

$YX = 4$

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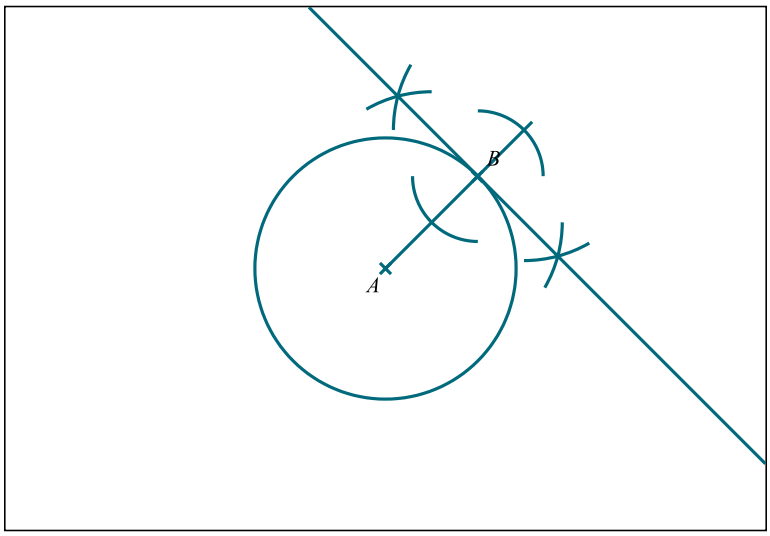
Question 21 of 48



Question 22 of 48

(a) $m\angle EDF = 53°$

(b) $m\angle DCE = 106°$

Question 23 of 48

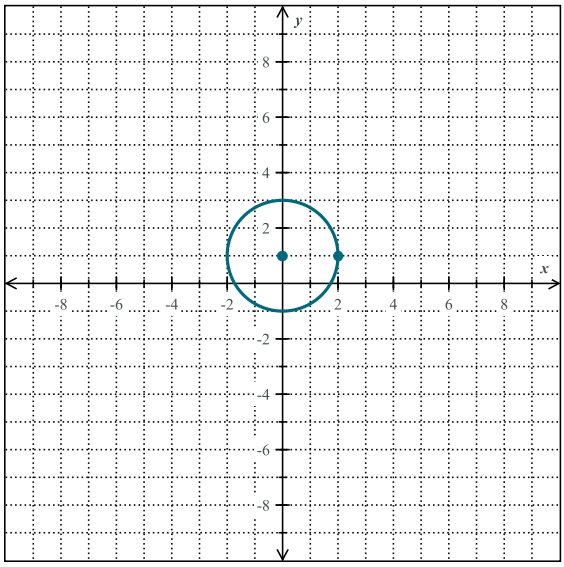
(a) $m\overset{\frown}{ST} = 43°$

(b) $m\overset{\frown}{AB} = 173°$

Question 24 of 48

Radius: 2

Center: $(0,\ 1)$



Question 25 of 48

$x^2 + y^2 = 16$

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Question 26 of 48

(a) Give the equation of the circle.

$$x^2 + y^2 = 16$$

(b) For each point, decide whether or not it is on the circle.

(x, y)	Is the point on the circle?	
	Yes	No
$(-1, \sqrt{7})$	○	◉
$(0, 5)$	○	◉
$(-4, 0)$	◉	○
$(\sqrt{5}, 2)$	○	◉

Question 27 of 48

$$(x+2)^2 + (y+3)^2 = 121$$

Question 28 of 48

(a) Find the following.

Center: $(5, 6)$

Radius: 4 units

Value of a: $x - 5$

Value of b: $y - 6$

(b) Use the Pythagorean Theorem to write an equation relating the side lengths of the right triangle. Write your answer in terms of x and y (with no other letters).

$$(x - 5)^2 + (y - 6)^2 = (4)^2$$

Question 29 of 48

$$x^2 + (y-2)^2 = 34$$

Question 30 of 48

$$(x+3)^2 + \left(y + \frac{1}{2}\right)^2 = \frac{25}{4}$$

Question 31 of 48

x-intercept(s): 0 , 2

y-intercept(s): 0

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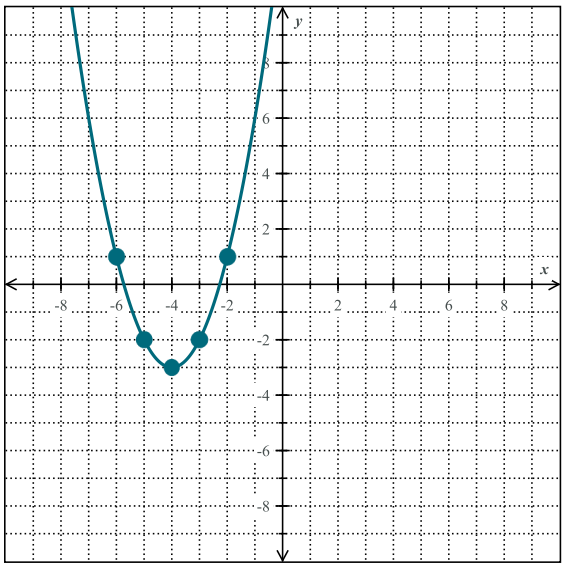
Question 32 of 48



Question 33 of 48

(a) Does the parabola open upward or downward?

○ upward ◉ downward

(b) Find the equation of the axis of symmetry.

equation of axis of symmetry: $x = -3$

(c) Find the intercept(s).

x-intercept(s): $-4, -2$

y-intercept(s): -8

(d) Find the coordinates of the vertex.

vertex: $(-3, 1)$

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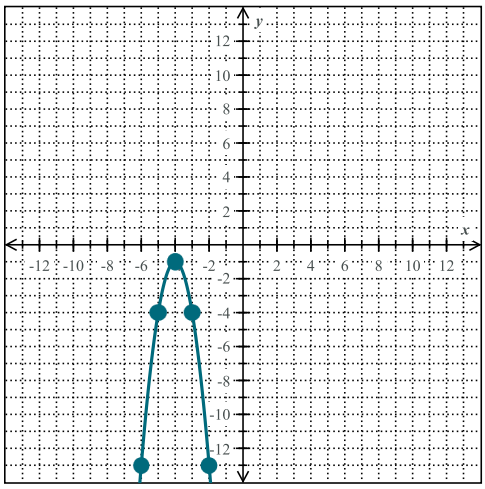
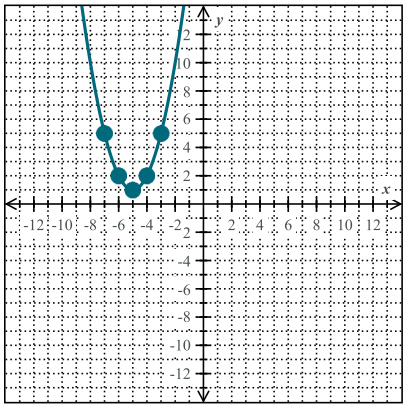
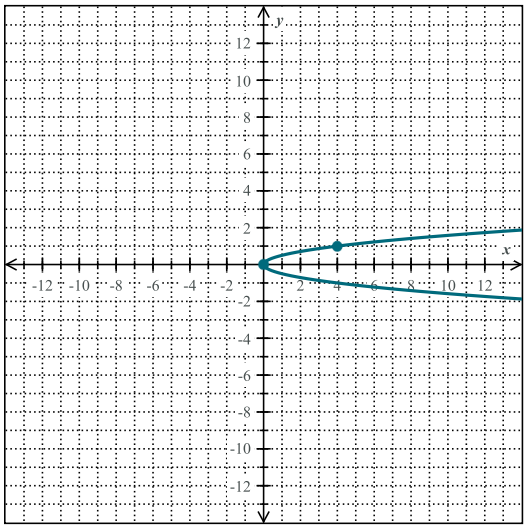
Question 34 of 48



Question 35 of 48



Question 36 of 48



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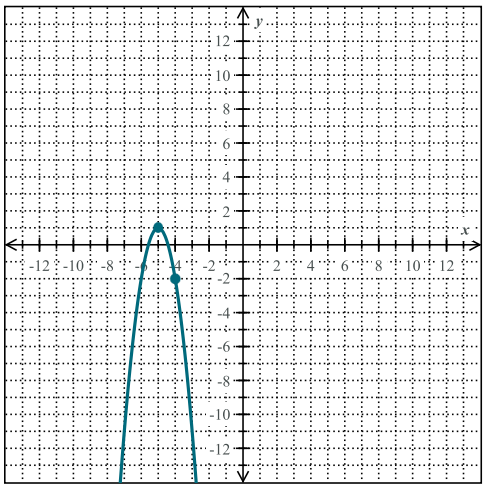
Question 37 of 48



Question 38 of 48

(a) What are the coordinates for the point T ?

$(-2, y)$

(b) Write the expressions for the distances FR and TR.

$FR = \sqrt{(x-2)^2 + y^2}$

$TR = x + 2$

(c) From the definition of a parabola, which of the following *must* be true?

$FR = TR$

(d) Use the above to write an equation for the given parabola solved for y^2.

$y^2 = 8x$

Question 39 of 48

y-intercept(s): 4

x-intercept(s): None

Question 40 of 48

$64v^3$

Question 41 of 48

$8u - 5$

Question 42 of 48

$w^2 - 7w - 8$

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Question 43 of 48

$3a^2 - 19a + 6$

Question 44 of 48

$4 - 12u + 9u^2$

Question 45 of 48

(a) Does the parabola open upward or downward?

◉ upward ○ downward

(b) Find the equation of the axis of symmetry.

equation of axis of symmetry: $x = 2$

(c) Find the coordinates of the vertex.

vertex: $(2, 1)$

(d) Find the intercept(s).

x-intercept(s): Undefined

y-intercept(s): 5

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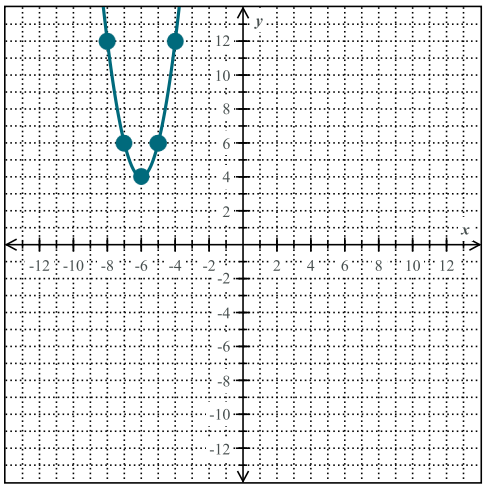

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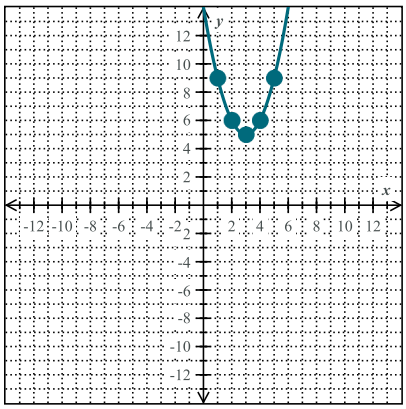

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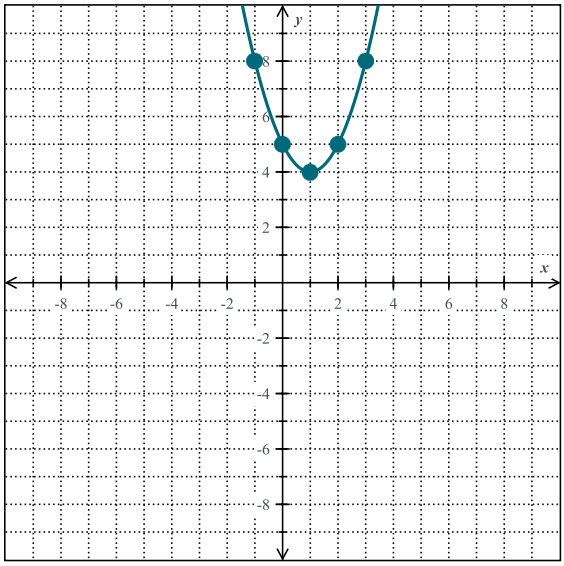

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